

Intermediate Macro

Chapter 16: Expectations, Output, and Policy

Instructor: Mihyun You

Department of Economics
Georgetown University

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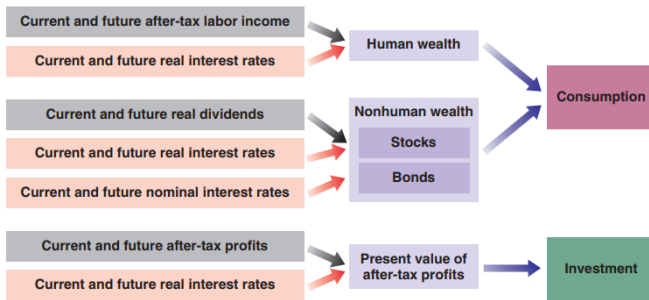
Key Tools and Concepts

- Chapter 16 introduces **rational expectations** in the context of the $IS - LM$ model by allowing economic agents to forecast the effects of future policy and to use there forecasts when determining current consumption and investment.

Expectations, Consumption, and Investment Decisions

16.1 Expectations and Decisions: Taking Stock

- Let us start by reviewing what we have learned, and then discuss how we should modify the characterization of goods and financial markets—the $IS - LM$ model—developed in the core.



Expectations and the IS Relation

16.1 Expectations and Decisions: Taking Stock

- Think about time in terms of two periods: the present and the future, which lumps all future years together.
- In Chapter 6 (equation (6.5)), we wrote the following equation for the IS relation:

$$Y = C(Y - T) + I(Y, r + x) + G$$

- We now want to modify this to take into account how expectations affect both consumption and investment.
- We proceed in two steps:
 - ① First, we rewrite the equation in more compact form without changing its content.

Aggregate Private Spending

16.1 Expectations and Decisions: Taking Stock

- For that purpose, let us define **aggregate private spending** as the sum of consumption and investment spending.

$$A(Y, T, r, x) \equiv C(Y - T) + I(Y, r + x)$$

where A stands for aggregate private spending, or simply, **private spending**.

- With this notation we can rewrite the IS relation as

$$Y = A(Y, T, r, x) + G \quad (16.1)$$

(+, -, -, -)

Introducing Expectations into IS Relation

16.1 Expectations and Decisions: Taking Stock

- ② The second step is to modify equation (16.1) to take into account the role of expectations.
- With the focus on expectations, the natural extension of equation (16.1) is to allow spending to depend not only on current variables but also on their expected values in the future:

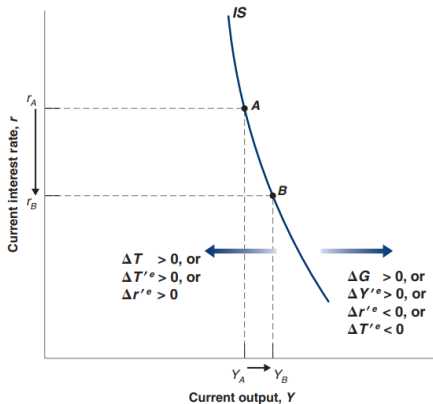
$$Y = A(Y, T, r, Y'^e, T'^e, r'^e) + G \quad (16.2)$$

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The New IS Curve

16.1 Expectations and Decisions: Taking Stock

- Given the values of expected future variables, the new IS relation remains downward-sloping in $Y - r$ plane, but it is likely to be steeper than the IS relation developed earlier.



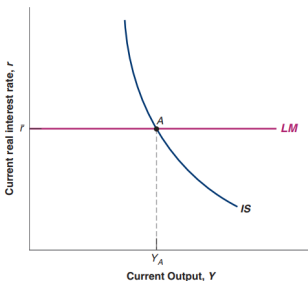
The New $IS - LM$

16.2 Monetary Policy, Expectations, and Output

- The LM relation is unaffected by the introduction of expectations, since money demand depends on the current level of transactions.
- The IS and LM relations are thus given by:

$$IS : Y = A(Y, T, r, Y'^e, T'^e, r'^e) + G \quad (16.3)$$

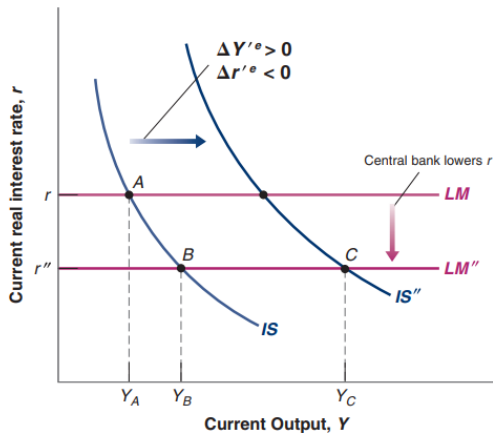
$$LM : r = \bar{r} \quad (16.4)$$



Monetary Policy Revisited

16.2 Monetary Policy, Expectations, and Output

- Now suppose the economy is in recession and the Fed decides to lower the real policy rate.



Role of Expectations on Policy Effects

16.2 Monetary Policy, Expectations, and Output

- The effects of monetary policy—the effects of any type of macroeconomic policy, for that matter—depend crucially on its effect on expectations:
 - If a monetary policy expansion leads financial investors, firms, and consumers to revise their expectations of future interest rates and output, then the effects of the monetary expansion on output may be large.
 - But if expectations remain unchanged, the effects of monetary expansion on output will be limited.
- Saying that the effect of a particular policy depends on its effect on expectations is not the same as saying that anything can happen.

Evolution of Expectations

16.2 Monetary Policy, Expectations, and Output

- Expectations are not arbitrary. Economists refer to expectations formed in the forward-looking manner as **rational expectations**.
- The last 40 years in macroeconomic research are often called the "*rational expectation revolution*".
- Expectations was referred to as **animal spirits** by Keynes in the *General Theory*.
- Economists have also assumed that people form **static expectations** (people expect the future to be like the present), and **adaptive expectations** (people "adapt" by revising their expectations over time).

Rational Expectations

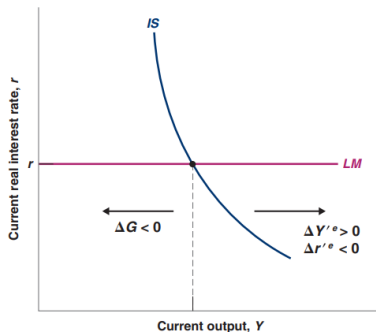
16.2 Monetary Policy, Expectations, and Output

- A group of macroeconomists led by Robert Lucas and Thomas Sargent argued that people look into the future and do the best job they can in predicting it.
- This is not the same as assuming that people know the future, but rather that they use the information they have in the best possible way.
- The introduction of the assumption of rational expectations, starting in the 1970s, has largely shaped the way macroeconomists think about policy.

Deficit Reduction Revisited

16.3 Deficit Reduction, Expectations, and Output

- In the basic $IS - LM$ model introduced in the Core, a reduction in the government budget deficit reduces current output.
- Once expectations are introduced, the effect of deficit reduction on current output becomes ambiguous.



The Effects of a Deficit Reduction on Current Output

16.3 Deficit Reduction, Expectations, and Output

- In response to the announcement of the deficit reduction, there are three factors shifting the IS curve:
 - ① Current government spending G goes down, leading the IS curve to shift to the left.
 - ② Expected future output Y'^e goes up, leading the IS curve to shift to the right.
 - ③ The expected future interest rate r'^e goes down,
- What is the net effect of these three shift in the IS curve?
- Without much more information about the exact form of the IS relation and about the details of the deficit reduction program, we cannot tell which shifts will dominate and whether output will go up or down.

Summary

16.3 Deficit Reduction, Expectations, and Output

- A program of deficit reduction may increase output even in the short run. Whether it does or does not depends on many factors:
 - ① The credibility of the program: Will spending be cut or taxes increased in the future as announced?
 - ② The composition of the program: Does the program remove some of the distortions in the economy?
 - ③ The state of government finances in the first place: How large is the initial deficit? Is this a "last chance" program? What will happen if it fails?
 - ④ Monetary and other policies: Will they help offset the direct adverse effect on demand in the short run?